

From fossil pollen to past vegetation

Can we help identify
microscopic fossils
at collection scale?

North American vegetation
~50 million years ago



Arecipites



Bombacacidites

Real specimens • expert annotation labels

Our task: find specimens, identify their category, support expert review.

Rich images. Uneven category coverage.

83

annotated slides

2,290

target annotations

32

candidate target categories

Multiple focal depths per slide

Read small image regions on Rice's cluster;
preserve the original labels and locations.

Bisaccate

316

Cupuliferoipollenites sp.

27

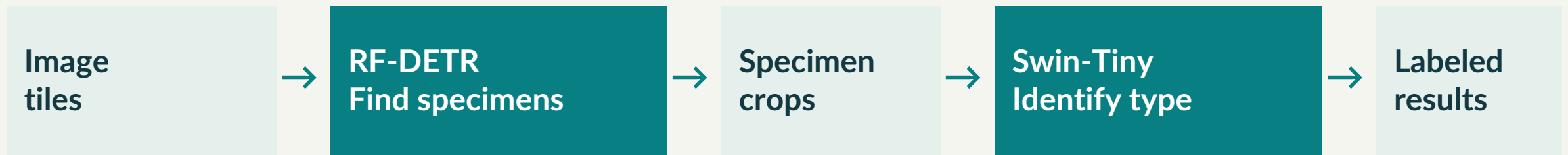
11,275 named annotations overall; 2,290 are in the initial target groups.

One joint model—or two specialized stages?

A / JOINT DETECTION + IDENTIFICATION



B / DETECTION FOLLOWED BY CLASSIFICATION



Same categories, slide split, preprocessing, and evaluation rules.

Existing detection models run on our images

64 known targets • 29 slides • two frozen YOLO checkpoints

54 / 64

Best focal plane

51 / 64

Focus stack

Target recovery at confidence ≥ 0.50 and box overlap (IoU) ≥ 0.50

Feasibility evidence—not classification accuracy or an independent test.

Measure biological recognition—not slide familiarity

01

Separate source slides

Keep each specimen and all its focal views in one data partition.

02

Evaluate every category

Use macro-F1, balanced accuracy, and per-category error analysis.

03

Test the full workflow

Count missed specimens as well as incorrect identifications.

Training-only balancing; fixed validation and test sets.

Build a useful, inspectable classification pipeline

1

Prepare

Inspect crops and agree on category groups.

2

Compare

Freeze the split and run both baselines.

3

Deliver

Show the specimen, proposed label, and uncertainty.

Goal: faster scientific review, with evidence about where models fail.